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BirdDreams

Joint Perspective and BEV Video Generation for a Plausible, Reactive Driving World Model

Joint Perspective & BEV Video Generation: A Plausible, Reactive Driving World Model

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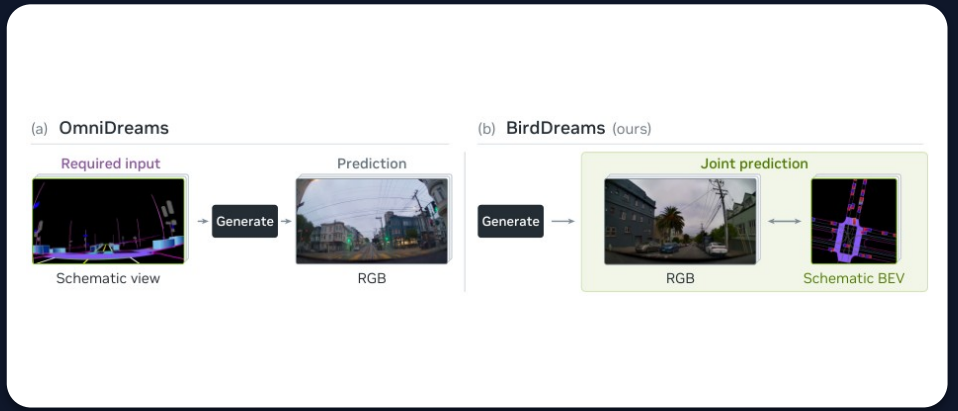
NVIDIA • KAIST | First submitted 2026-11-10



Source: paper Fig. 1

In short: layout is jointly predicted, not given as input

BirdDreams maps, via two joint-attn. causal DiTs, **one front image** to front-view + metric **BEV videos**; latest BEV frame, **shifted + rotated** per ego path = soft action cue → better control, plausible counterfactual reactions.

−35% / −36%

ADE with ego-action conditioning
vs. OmniDreams (Plücker)
PhysicalAI-AV / nuScenes

−21% / −37%

minADE without action conditioning
vs. OmniDreams (Vanilla), same data
PhysicalAI-AV / nuScenes

57% : 24%

Human preference: more realistic agents
BirdDreams : baseline (plus 19% ties)
25 raters, 375 pairwise judgments

The dilemma: other agents either replay logs or are hard to learn

Layout-conditioned

E.g., OmniDreams: needs HD maps + future agent trajectories as input

Log replay: agents ignore ego deviations from the log

External traffic model: interaction lies outside the world model

Perspective-only video

Motion entangled with texture and lighting

Vehicles often occluded

Same pixel shift ≠ same speed (perspective scaling)

Layout-free baselines lose vehicles or act erratically at complex intersections

BirdDreams' Idea

Turn layout from an input into a jointly predicted output

Inference needs only one front-view image

No HD map or vehicle states needed

Source: paper Fig. 1

Why schematic BEV: it complements perspective RGB

Schematic BEV: good at dynamics

- No texture or lighting: simpler target
- Fixed metric scale: pixel shift = same distance
- Top-down view: little inter-vehicle occlusion
- Red front, blue rear: heading at a glance
- Clear vehicle–road and lane relations

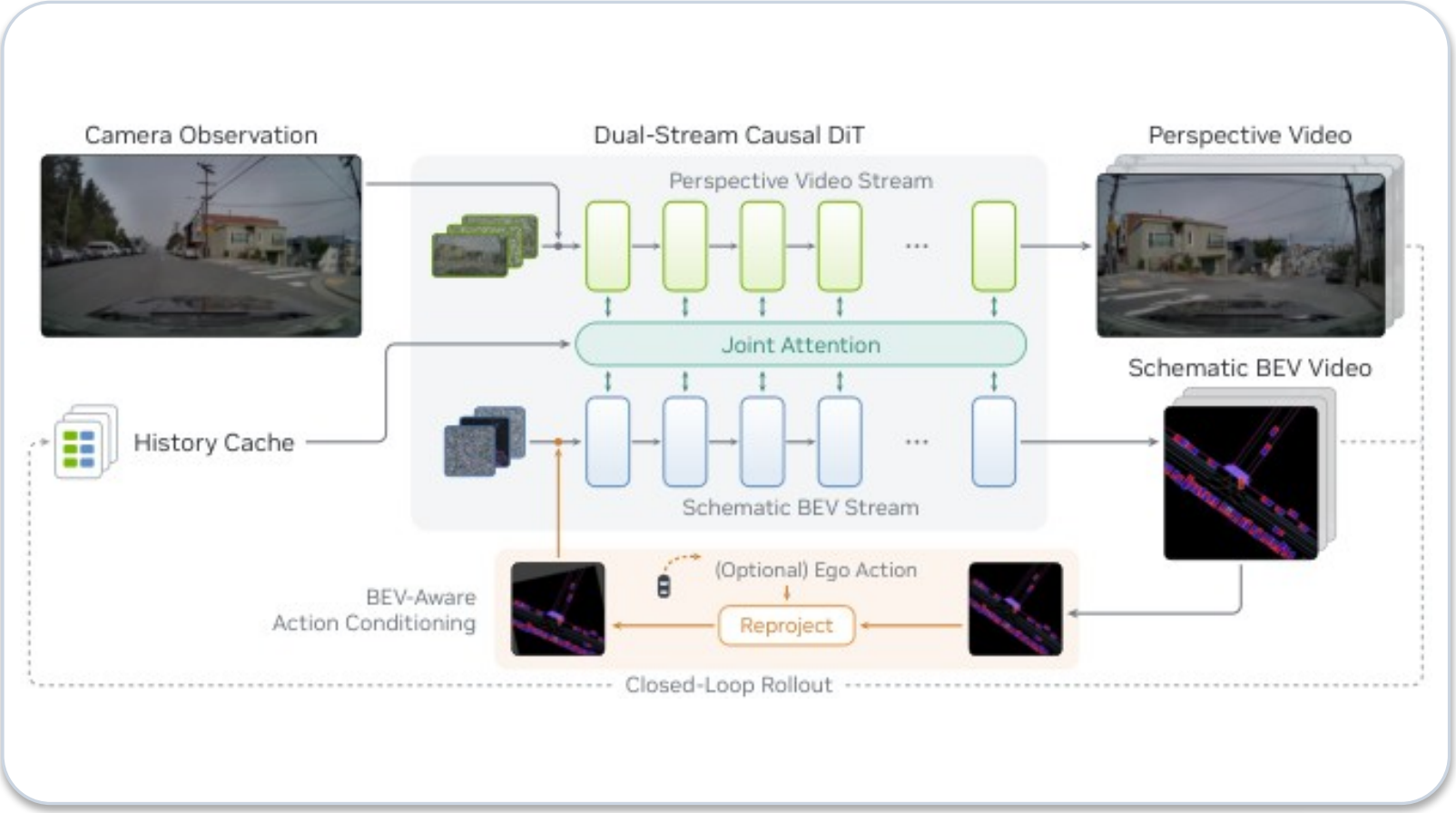
Joint Attention

Perspective RGB: good at appearance

- Traffic signals, e.g., light states
- Road details and object appearance
- Distant objects beyond BEV range (~69 m)
- The very observation fed to simulation

BEV rendering Top-down ortho., ego-centered, heading up • ~69.3 m × 69.3 m/frame • road edges magenta, crosswalks purple, pedestrians yellow • oriented vehicle boxes • trained on renders of maps + 3D boxes (interp. fills frames/occlusions; dedup.)

Architecture: dual-stream causal DiT with joint attention in every block



Inference: generate B_0 from I_0, then jointly generate both videos chunk by chunk; after each chunk, cache its KV and update the BEV reference to its last frame

Weight-shared init.

Perspective branch = pretrained vanilla OmniDreams

BEV branch = copy of the same weights

Joint attn. + separate experts

Each block's Q attends to both streams' K/V

Separate projections, FFNs, output heads

Block-causal AR

One chunk = 3 latent frames

Bidirectional within, causal across chunks; KV cache

≈4B params · RGB 848×480 · BEV 480×480

Source: paper Fig. 2; details in § 3.2.1, § 4.1

Per-stream noise: cross-stream cues and flexible conditioning

$$x = (1 - \sigma) \cdot z + \sigma \cdot \varepsilon$$

σ sampled independently across chunks and across the RGB / BEV streams

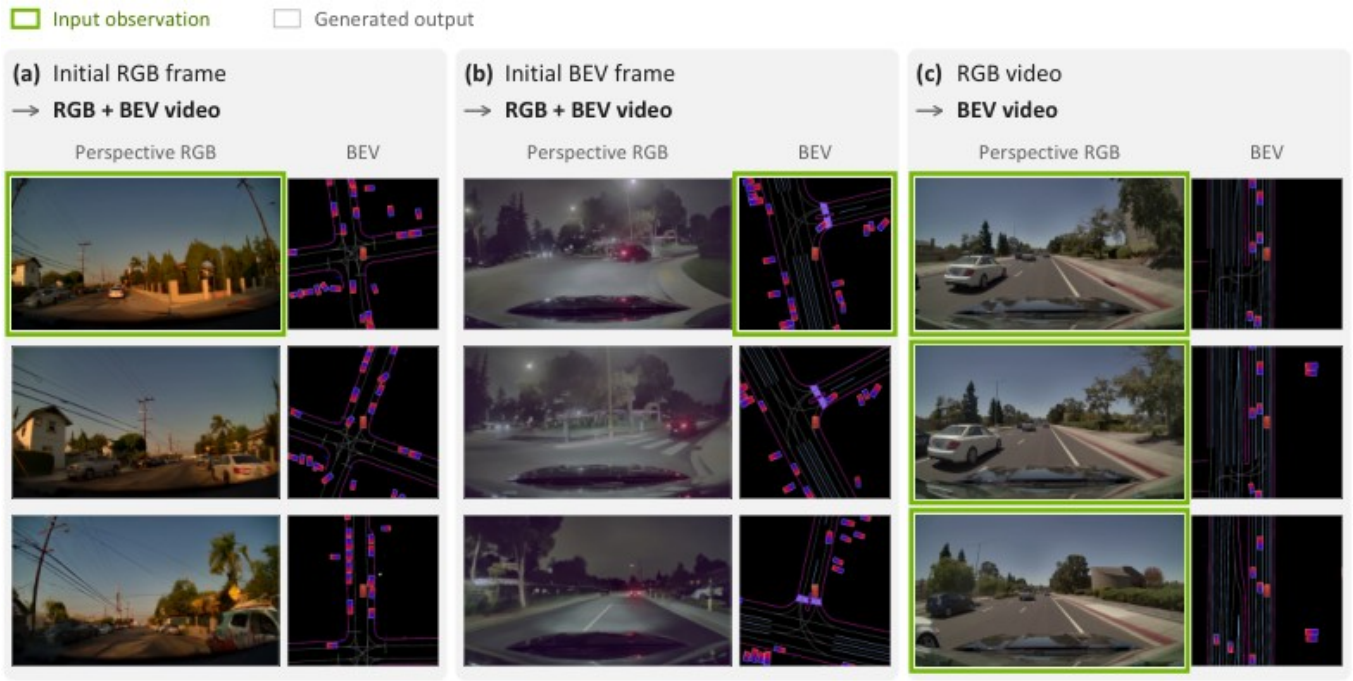
Clean BEV + noisy RGB
Road geometry, vehicle positions → guide RGB

Clean RGB + noisy BEV
Visual cues (e.g., lights) → complete BEV

Both noisy
Denoised jointly via joint attention

Loss: normalized by latent count within each stream, then streams weighted equally so higher-res RGB does not dominate

Flex. cond.: one model denoises only unknown latents (green: input)



Source: paper Fig. 3; method in § 3.2.2, § 3.2.5

Actions: last BEV frame rigidly warped by ego path as a soft condition

① Reference
Last generated BEV frame

B_r of prior chunk (B_0 first)

② Rigid reproj.
Per pose (p_t, θ_t): translate + rotate, Eq. (4)

③ Encode + noise
VAE encoding, small noise $\sigma_cond = 0.1$

④ Soft injection
Zero-init projection added to BEV input tokens

⑤ Cross-stream
Joint attention to RGB branch → generates RGB

$$\tilde{B}_t(u) = B_r(\Pi[R(\theta_r)^T (R(\theta_t) \cdot \Pi^{-1}(u) + p_t - p_r)])$$

Π: ego coords (m) → BEV pixels; R(θ): 2D rotation. Meaning: how the scene would move if only the ego moved and all else were static

Why it works

Orthographic + metric: translation = uniform pixel shift

Turning = rotation about ego

No depth estimation (vs. Gen3C warp rendering)

Why "soft"

Reprojection shows only ego-induced motion

Supervised by the true future BEV

Model fills in agent motion and new content

Closed loop, optional

Reference updated per chunk to reflect newly predicted scene

Condition dropout in training

→ same model also does action-free rollouts

Training: Diffusion Forcing, then Resampling Forcing vs. drift

Stage 1 • Diffusion Forcing

History = noised ground-truth chunks

Noise sampled independently across chunks and RGB/BEV streams

Flow-matching loss, velocity target $\varepsilon - z$

Stage 2 • Resampling Forcing fine-tuning

Shared noise τ corrupts GT → one-step Euler AR denoising → predicted history (grad. stopped)

Exposes the model to errors in its own history

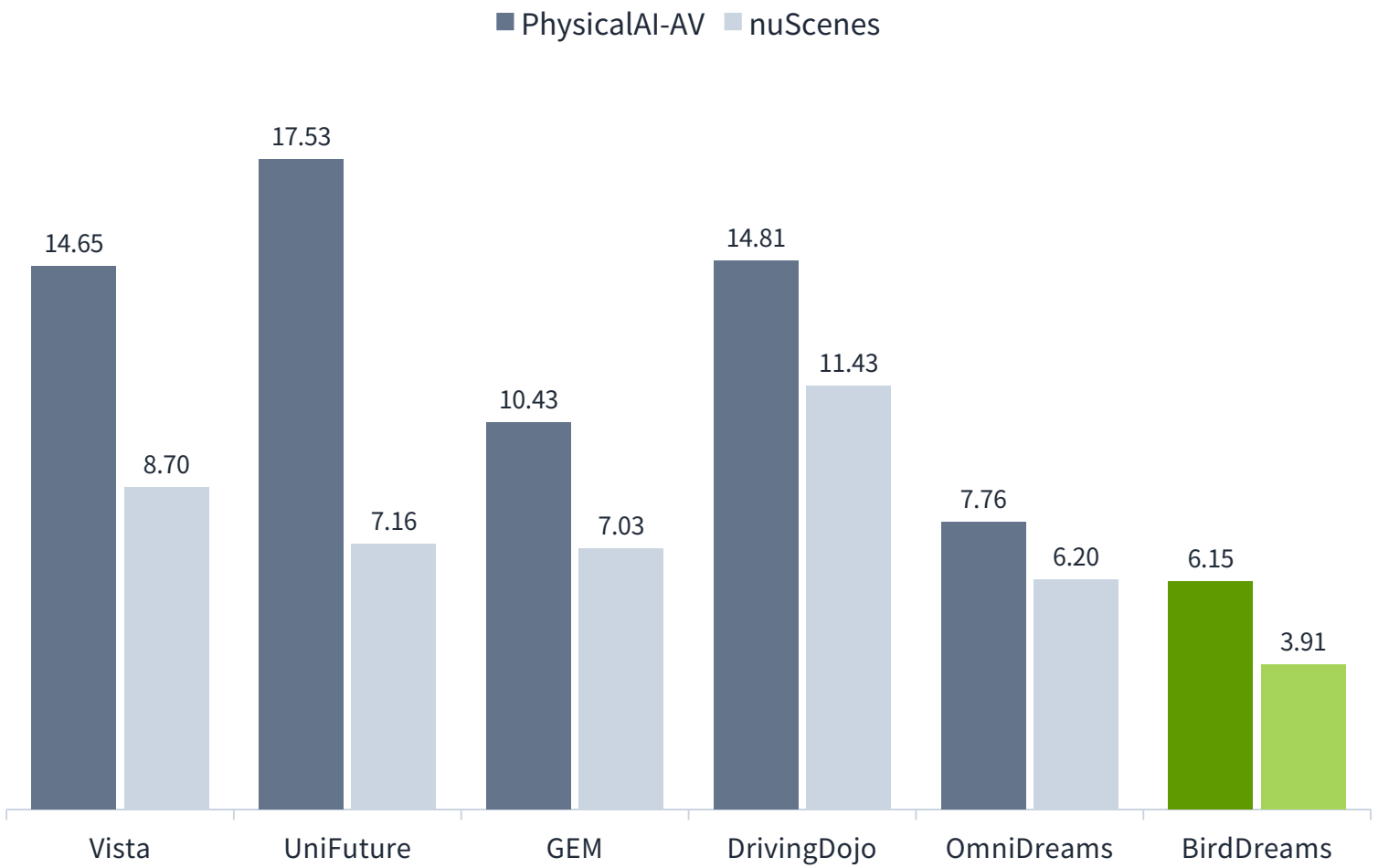
Target chunks still noised independently; parallel training

Data & compute RDS-HQ-1M subset ~670 h • nuScenes train ~3.9 h (post-trained for eval.) • 64×H100 • AdamW, lr 3e-5

Source: paper Fig. 4; details in § 3.2.4, § 3.3, § 4.1

Without actions: BirdDreams' ego motion is closest to real logs

minADE ↓ (m, min over 3 samples) OmniDreams = Vanilla version



−21% / −37%

minADE vs OmniDreams (Vanilla)

4.15 → 0.43

nuScenes FTD (trajectory distribution distance)

Also best video quality

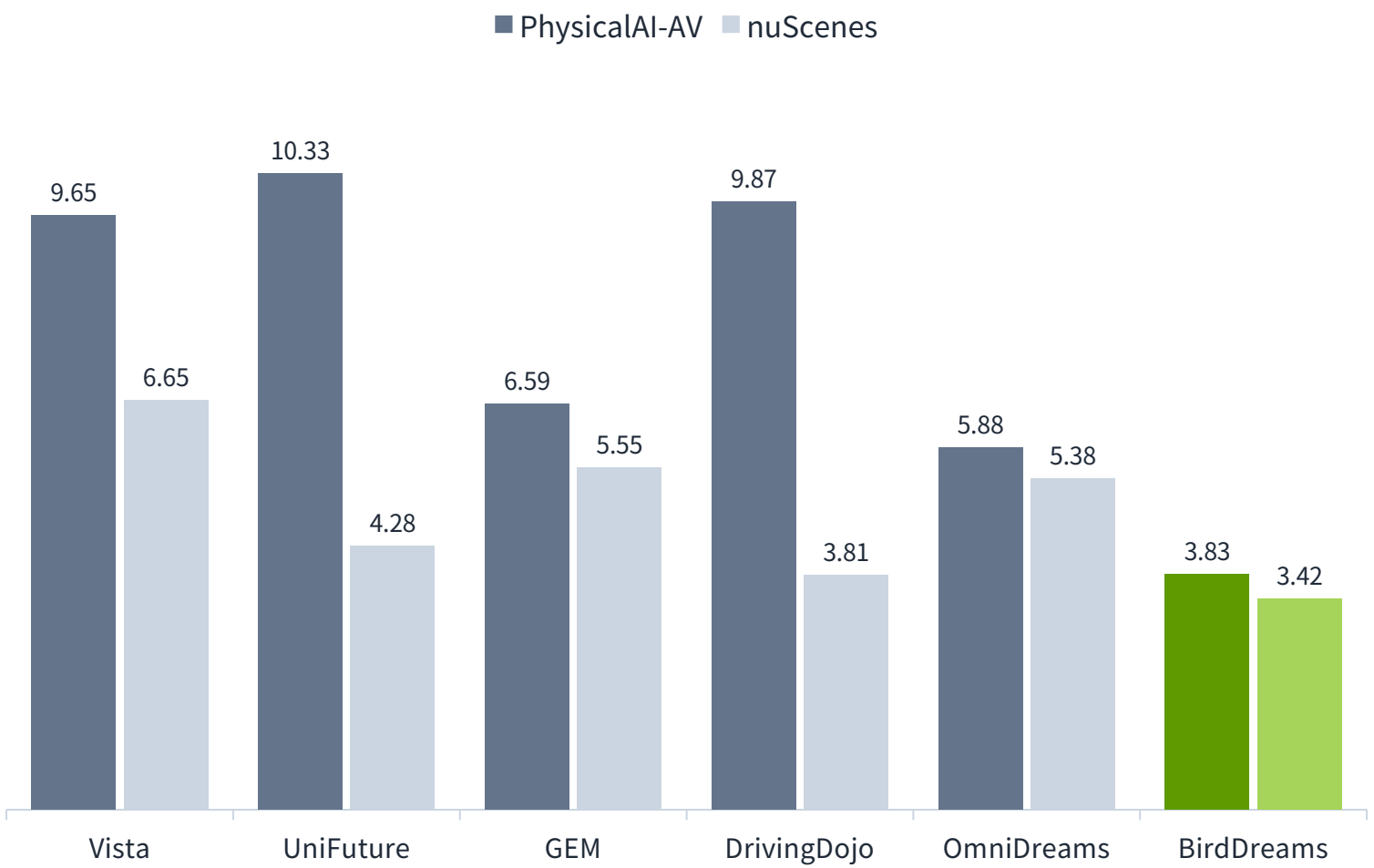
FVD 271 / 309, FID 10.70 / 11.48 (both lowest)

Image Quality 0.428 / 0.439 only 2nd; gains mainly in motion

Data: paper Tab. 1 (each cell: PhysicalAI-AV / nuScenes)

With actions: BirdDreams follows trajectories best on both datasets

ADE ↓ (m, single sample) OmniDreams = Plücker version



−35% / −36%

ADE vs OmniDreams (Plücker)

vs. OmniDreams (Plücker)

DTW 197.1 → 95.2 / 234.8 → 133.3
FTD 3.02 → 0.48 / 1.47 → 0.56
FVD 271.5 → 238.9 / 343.1 → 294.0
FID 15.35 → 11.81 / 18.27 → 14.80

Note

This comparison changes both BEV prediction and action injection; the paper does not ablate them separately

Data: paper Tab. 2 (each cell: PhysicalAI-AV / nuScenes)

Agents react: when the ego changes behavior, others brake and yield

(a) No action

Ego and crossing car yield; both stop

Ego motion predicted by the model

(b) Ego forces through

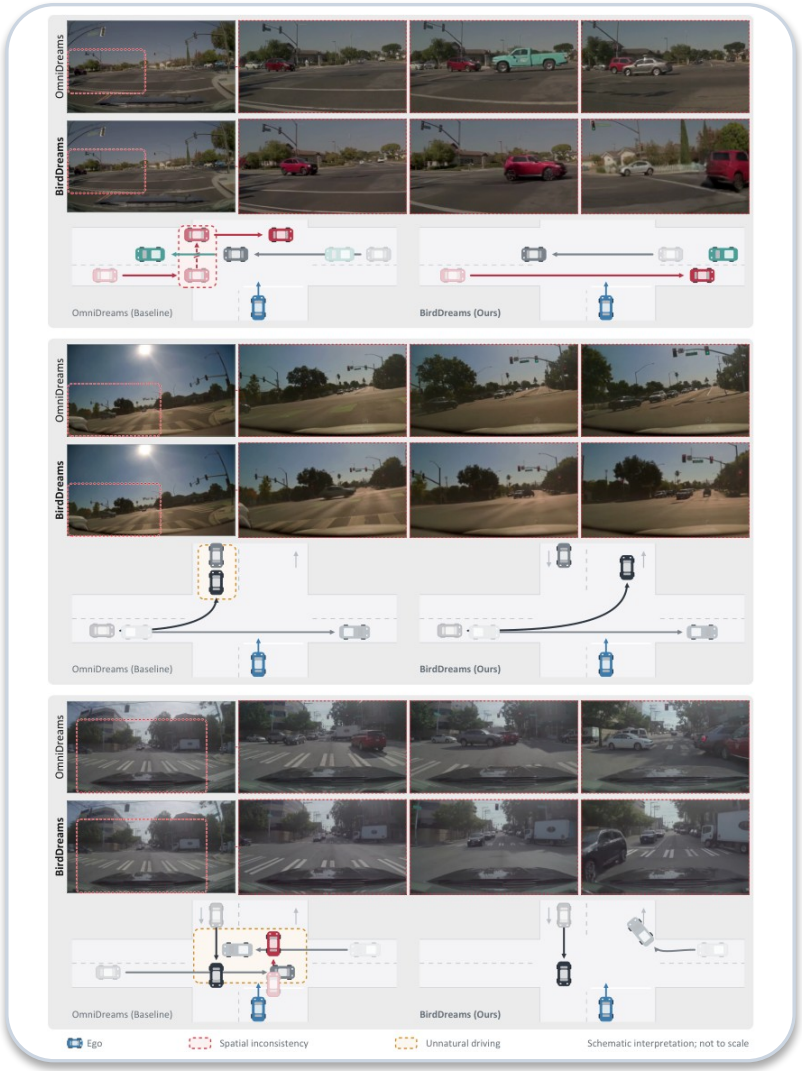
Crossing car stops to yield

(b) Ego cuts in

Rear car slows, keeps gap

Extreme, unavoidable actions cause crashes, deemed plausible by authors

Same road layout: RGB-only baselines often move implausibly



Comparison setup (for fairness)

OmniDreams (HDMap) vs. BirdDreams (HDMap): same initial observation, counterfactual ego action and static road geometry; no agent states given, so agent motion is fully model-predicted

Scenario 1 • Intersection crossing

Baseline red car "jumps" to the opposite lane (spatially inconsistent)

BirdDreams red car crosses smoothly within one lane

Scenario 2 • Left turn

Baseline turning car crowds into a same-lane vehicle (unnatural)

BirdDreams turning car enters the correct lane

Scenario 3 • Urban intersection

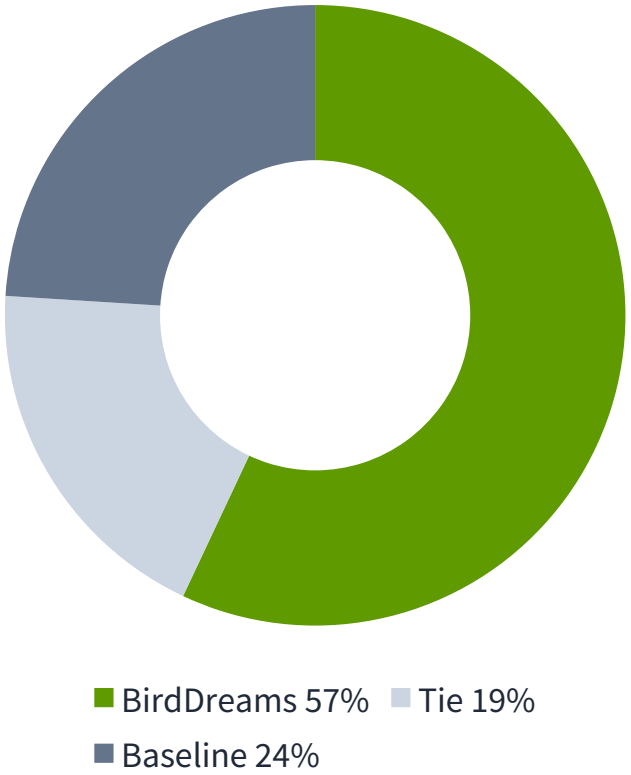
Baseline paths conflict at the center; incoherent interaction

BirdDreams oncoming car goes straight, side car turns; consistent

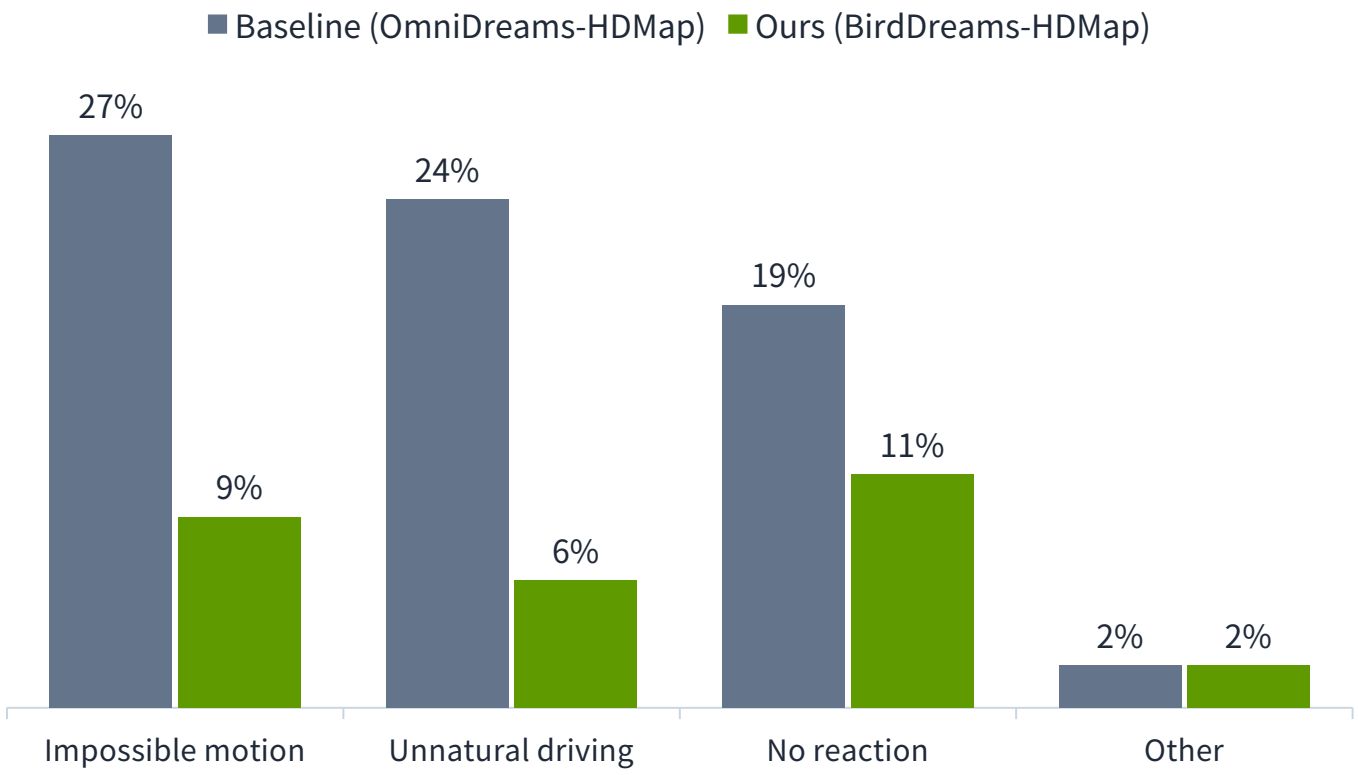
Source: paper Fig. 6; setup in § 4.3

Human study: 57% of judgments find BirdDreams' agents more realistic

(a) Overall preference



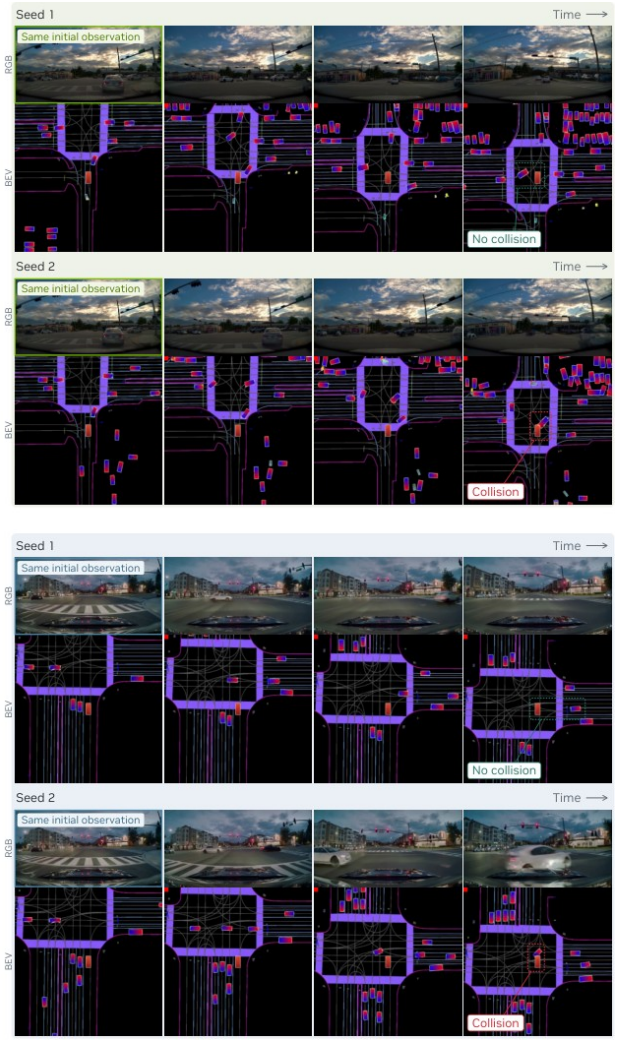
(b) Issues flagged for losing videos (% of all 375; multi-select)



Setup 25 raters × 15 random pairs (pool 119, shuffled); Q: more realistic agent behavior?; unavoidable crashes count as realistic

Data: paper Fig. 8 (N = 375 pairwise judgments, 25 participants)

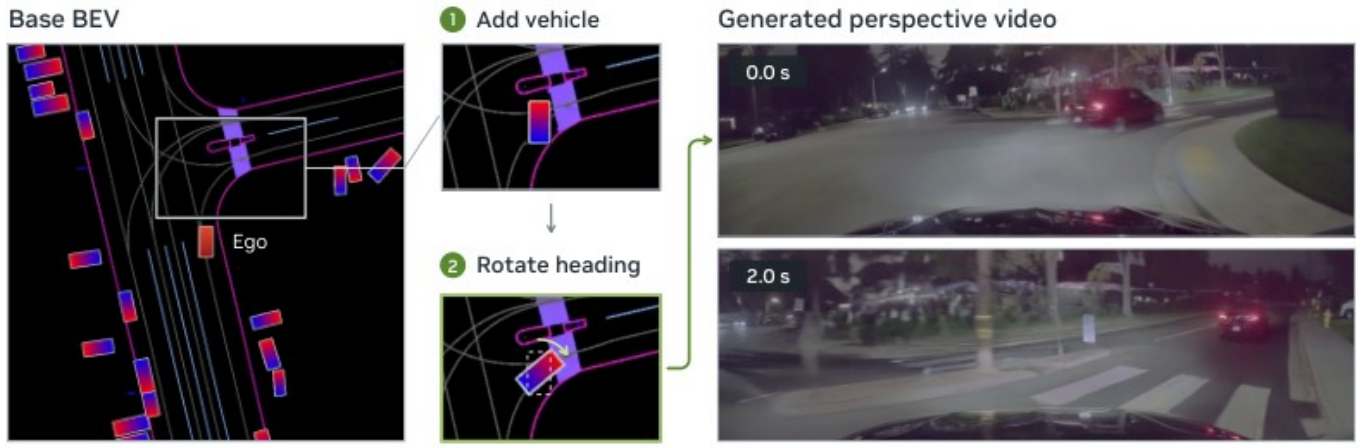
One action can yield different outcomes; scenes are editable in BEV



Diverse futures same initial observation, counterfactual trajectory and road geometry; only the seed changes:

Seed 1 → no collision **Seed 2 → collision**

The model samples different agent reactions to the same ego action

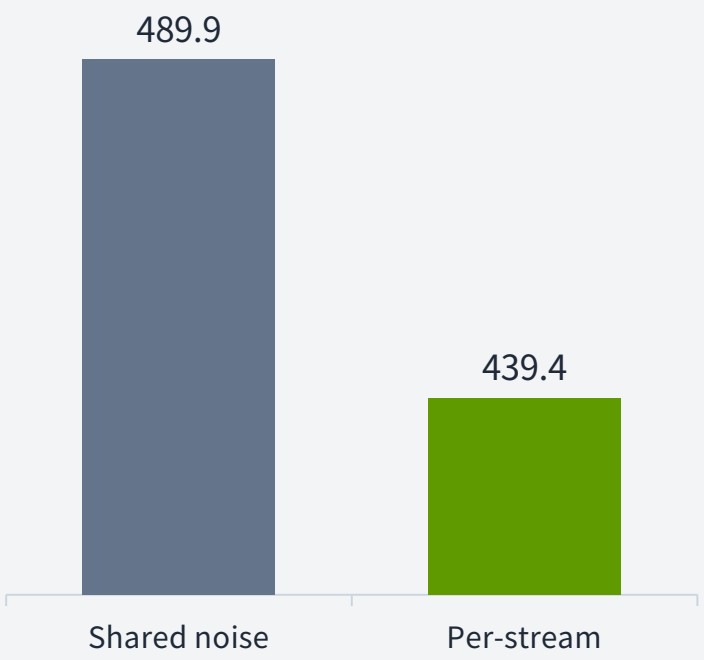


BEV scene editing add car → set heading → generate RGB video (incl. first frame) directly; no 3D box handling or real images

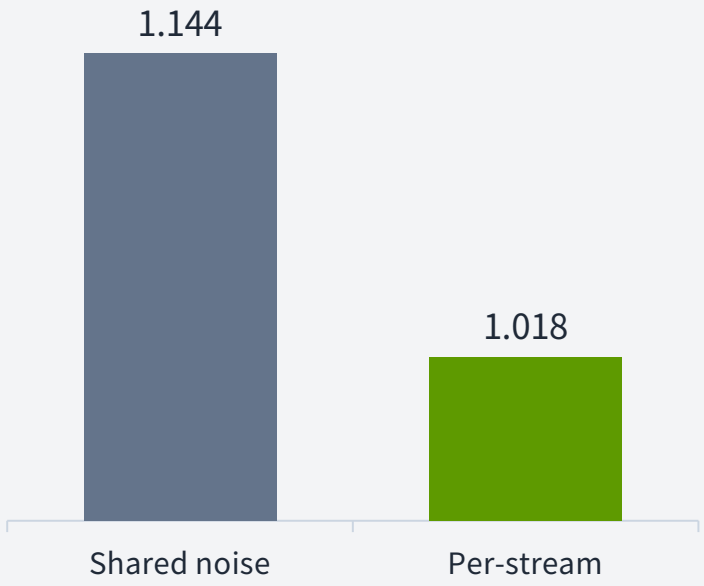
Source: paper Fig. 7, Fig. 11

Ablations/ext.: per-stream noise helps; BEV mildly aids traj. pred.

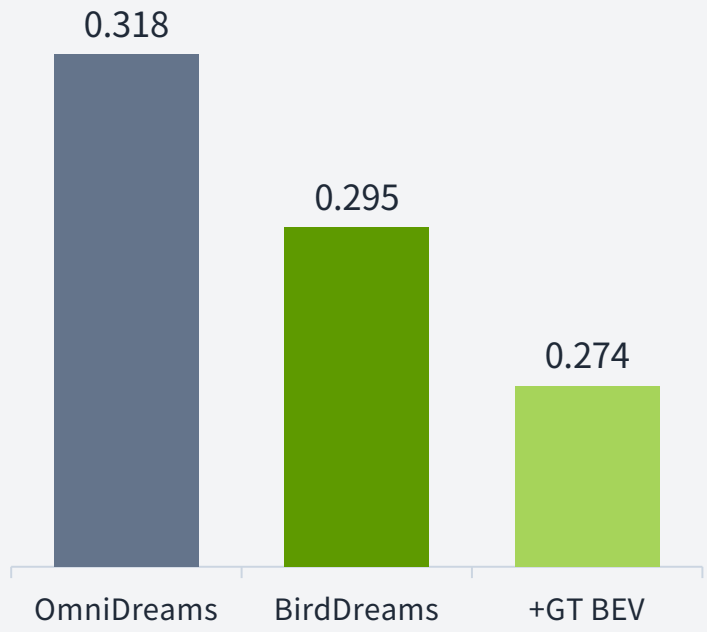
Noise sampling • FVD ↓
RDS-HQ-1M val. set



Noise sampling • FTD ↓
RDS-HQ-1M val. set



WAM • minADE@3s ↓ (m)
World-action model, 100 clips, best-of-6



Per-stream noise FVD −10%, FTD −11%: training with one clean and one noisy stream pays off

WAM −7.3% preliminary; +GT BEV uses privileged input

Data: paper Fig. 10 (§ 4.4.1, § 4.4.2)

Limitations & Open Questions

No reactivity metric

Key claim rests on qualitative examples and a 25-rater study, which used the HDMap variant given HD maps

Confounded action study

Tab. 2 changes both BEV prediction and action injection without separate ablations

Noisy eval. protocol

Trajectories reconstructed from video via Depth Anything 3; extra nuScenes post-training warrants caution

Label reliance

BEV supervision needs maps + 3D boxes, hard to scale to unlabeled video; covers only ~69 m; signals rely on RGB

Narrow setup

Single front camera; reprojection assumes a static reference; inference speed, key for closed loop, unreported

Preliminary WAM

Only 100 clips; minADE gain 0.023 m (7.3%); authors call it preliminary

Co-generated structured view: more controllable ego, plausible agents

BEV as supervision

A low-dim., metric, low-occlusion schematic view eases learning agent dynamics; preference 57% vs. 24%

BEV as interface

Rigid reprojection + soft conditioning executes metric paths: vs. OmniDreams (Plücker), ADE ~35% lower, DTW ~halved

BEV is editable

Per-stream noise enables any-stream conditioning: add cars and set headings in BEV to generate RGB scenes

Notable: a clear path to reactive closed-loop simulation without external traffic models; **Open:** systematic quantitative reactivity eval., multi-camera setups, and inference efficiency.